DT 111 · TOPIC 3

Exploratory Data Analysis

Descriptive statistics, visualisation, patterns and cautious interpretation

Mkwawa University College of Education

BSc Data Science and Artificial Intelligence

Learning outcomes

- Explain the purpose and iterative nature of EDA.
- Summarise numerical and categorical variables appropriately.
- Interpret centre, spread, quartiles and simple correlation.
- Choose suitable basic charts for analytical questions.
- Identify patterns while avoiding unsupported causal claims.

A small student dataset

Student	Attendance (%)	Final score	Region
A	92	82	Iringa
B	78	71	Mbeya
C	65	60	Iringa
D	88	79	Arusha
E	52	48	Mbeya

Initial inspection considers variable types, possible patterns and questions that deserve further analysis.

Exploratory Data Analysis (EDA)

EDA is disciplined curiosity, not simply “making charts”.

Inspect

Understand rows, columns, variable meanings and data quality.

Summarise

Use counts, centre and spread to describe the data.

Visualise

Use charts to reveal shape, groups, trends and relationships.

Interpret

Connect patterns to context and identify new questions.

EDA is iterative

1. **Question**
   What do we want to understand?
2. **Inspect**
   Structure and quality
3. **Summarise**
   Counts and descriptive statistics
4. **Visualise**
   Choose suitable charts
5. **Interpret**
   Patterns, anomalies and next questions

Core EDA questions

- How many observations and variables are there?
- What does each variable represent?
- Which variables are numerical or categorical?
- What is typical, and how much variation exists?
- Are groups different or variables related?
- Are any observations unusual or missing?

Frequency table for a categorical variable

Region	Count	Percentage
Iringa	2	40%
Mbeya	2	40%
Arusha	1	20%

Counts and percentages are natural first summaries for categorical variables.

Measures of centre

Scores: 48, 60, 71, 79, 82

Mean

68

(48+60+71+79+82) / 5

Median

71

Middle ordered value

Mode

None

No value repeats

Effect of an extreme value

The median is more resistant to an extreme value than the mean.

Original values

- 48, 60, 71, 79, 82
- Mean = 68
- Median = 71

One extreme value

- 48, 60, 71, 79, 200
- Mean = 91.6
- Median = 71

Centre is not enough

Centre

What value is typical?

Spread

How variable are the observations?

Shape

How are values distributed?

Outliers

Are any values unusually far from the rest?

Measures of spread

At this level, interpretation is more important than memorising every formula.

Range

Maximum - minimum. Here: 82 - 48 = 34.

Interquartile range (IQR)

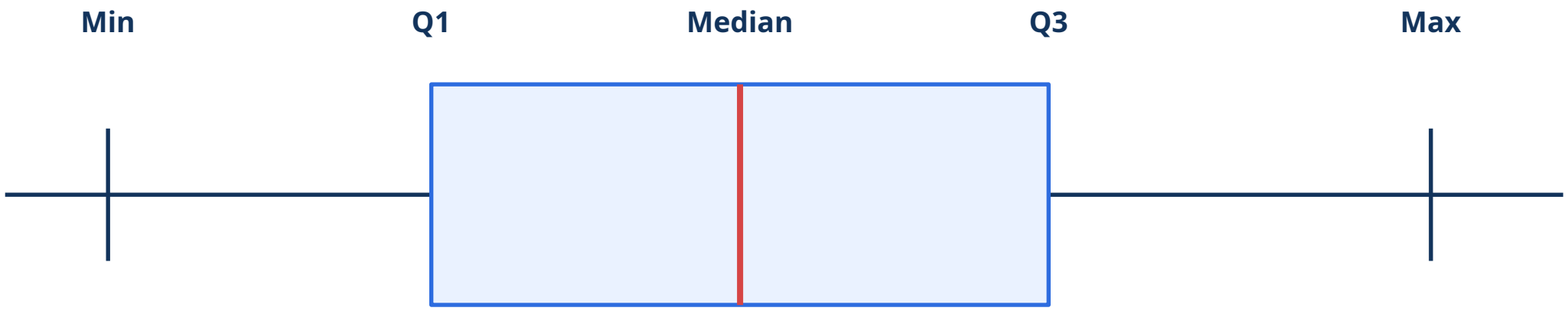
Q3 - Q1: spread of the middle 50%.

Standard deviation

Describes how far values typically lie from the mean.

Five-number summary and box plot

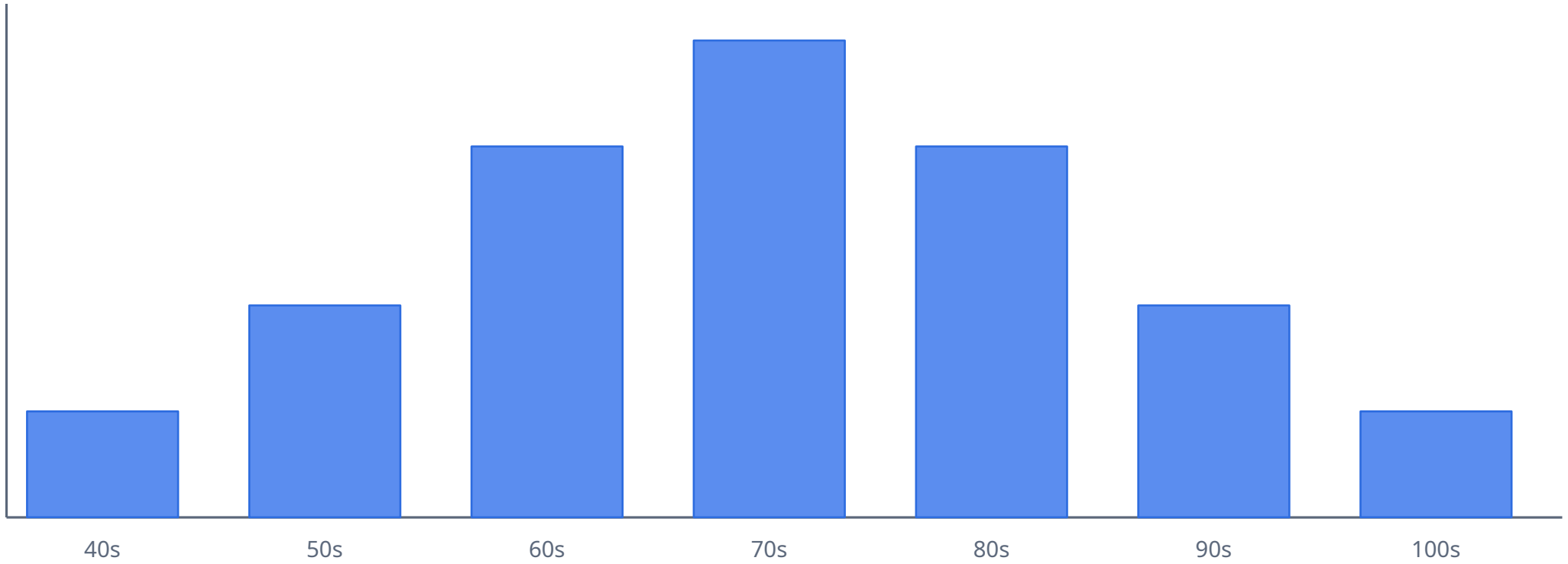
Minimum, Q1, median, Q3 and maximum provide a compact description of a numerical distribution.



The box contains the middle 50% of values. The median marks the centre of that middle half.

Distribution shape

A roughly symmetric distribution has similar left and right tails.



Skewed distributions

Right-skewed

Most values are lower, with a long tail toward high values. The mean is often pulled upward.

Left-skewed

Most values are higher, with a long tail toward low values. The mean is often pulled downward.

Why it matters

Distribution shape helps explain why mean and median may differ.

Bar chart and histogram

Bar chart

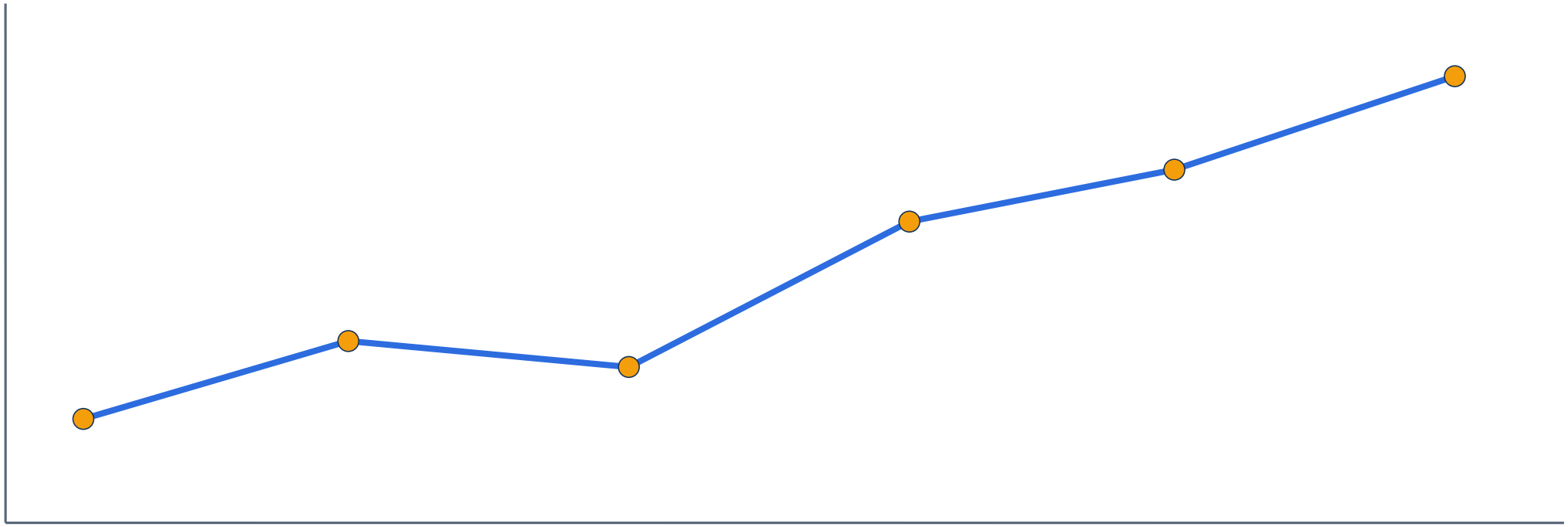
- Categorical variable
- Bars represent separate categories
- Gaps between categories are natural
- Example: students by region

Histogram

- Numerical variable
- Bars represent adjacent intervals
- Bars normally touch
- Example: distribution of final scores

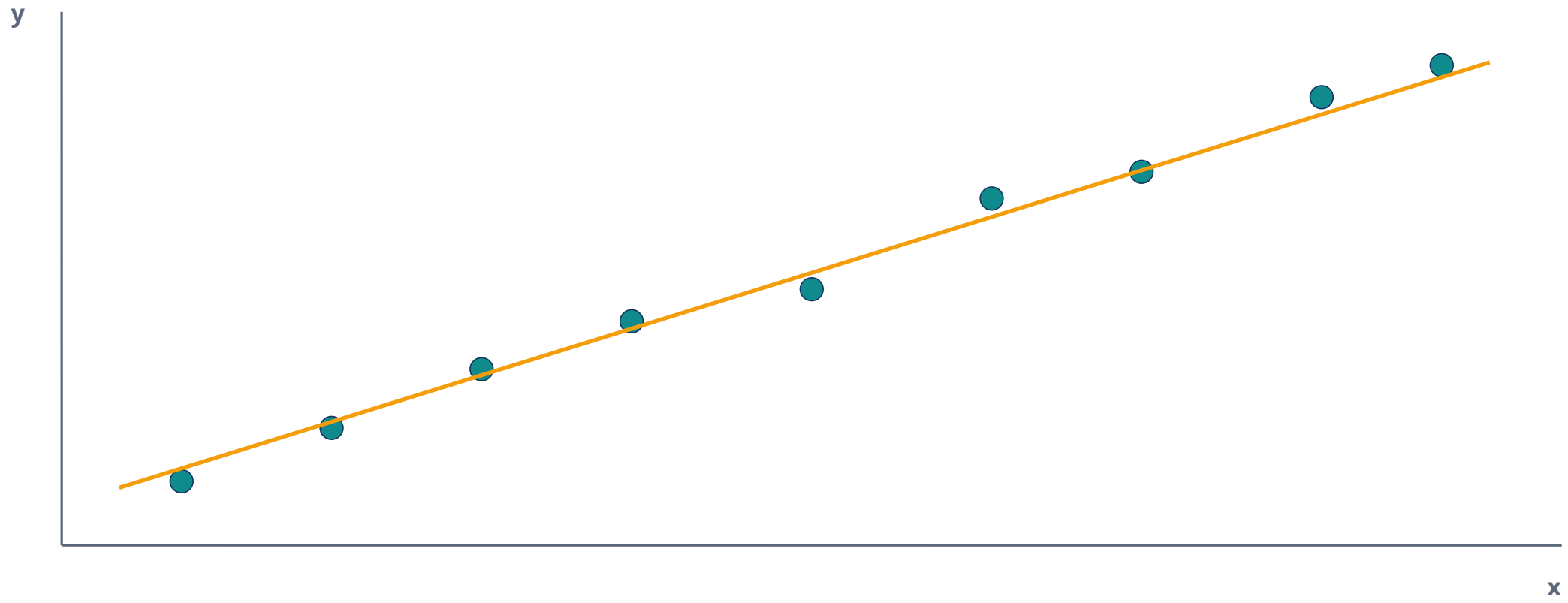
Line chart for ordered or time data

A line chart is useful when observations have a meaningful order, especially time.



Scatter plot for two numerical variables

Each point represents paired values, such as attendance and final score.



Correlation coefficient

Correlation summarises the direction and strength of a linear association.

Strong negative

-1

As one rises, the other tends to fall

No linear relationship

0

No consistent straight-line pattern

Strong positive

+1

Both tend to rise together

Patterns commonly examined in EDA

Distribution

Where values concentrate and how they spread.

Relationship

How two variables vary together.

Group difference

How categories compare.

Trend

How values change over an ordered dimension.

Anomaly

A value that differs noticeably from the general pattern.

Association is not causation

Observation

- Higher attendance tends to occur with higher scores
- This is evidence of a positive association

Unsupported conclusion

- “Attendance alone causes every score to increase”
- Other variables may influence both attendance and performance

A chart should answer a question

Weak

“I made a chart because a chart is required.”

Better

“I used a histogram to see how final scores are distributed.”

Strong interpretation

“Scores concentrate around 65-80, with a few low cases worth investigating.”

Mini-EDA: from table to insight

1. **Inspect**
   5 students, 4 variables
2. **Summarise**
   Mean 68, median 71
3. **Visualise**
   Attendance vs score
4. **Compare**
   Groups are small
5. **Investigate**
   Low score of 48
6. **Interpret**
   Association, not cause

Topic 3 key ideas

- EDA combines numerical summaries, visualisation and context.
- Centre, spread and shape describe different aspects of a distribution.
- Chart choice depends on the variable type and analytical question.
- Correlation describes association, not causation.
- A good EDA produces evidence-based observations and better questions.